

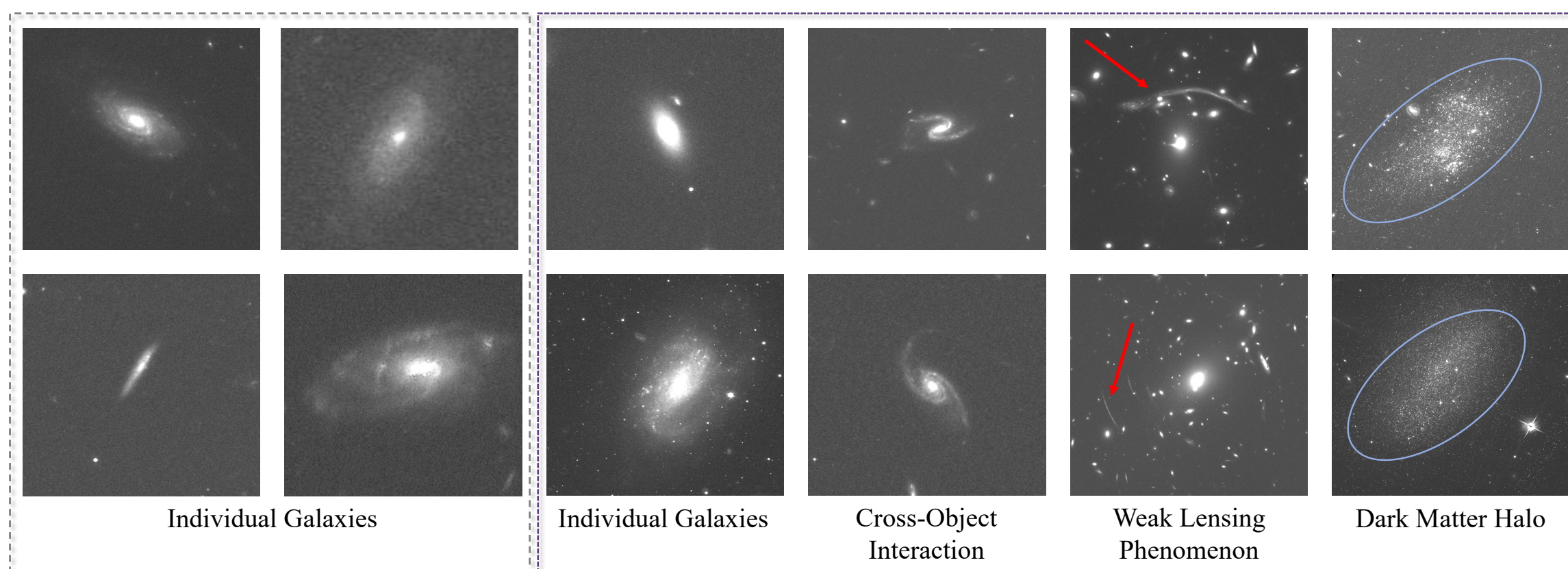
Problem Formulation

Task: Physically-Constrained Field-Level SR

Key Constraint: Reconstructions must preserve **Flux Consistency** (total object light intensity) —a physical principle violated by standard SR methods.

Purpose: Detecting distant faint celestial bodies, precise photo-metric measurements...

Motivation



(a) Previous Dataset Examples

(b) STAR Dataset Examples

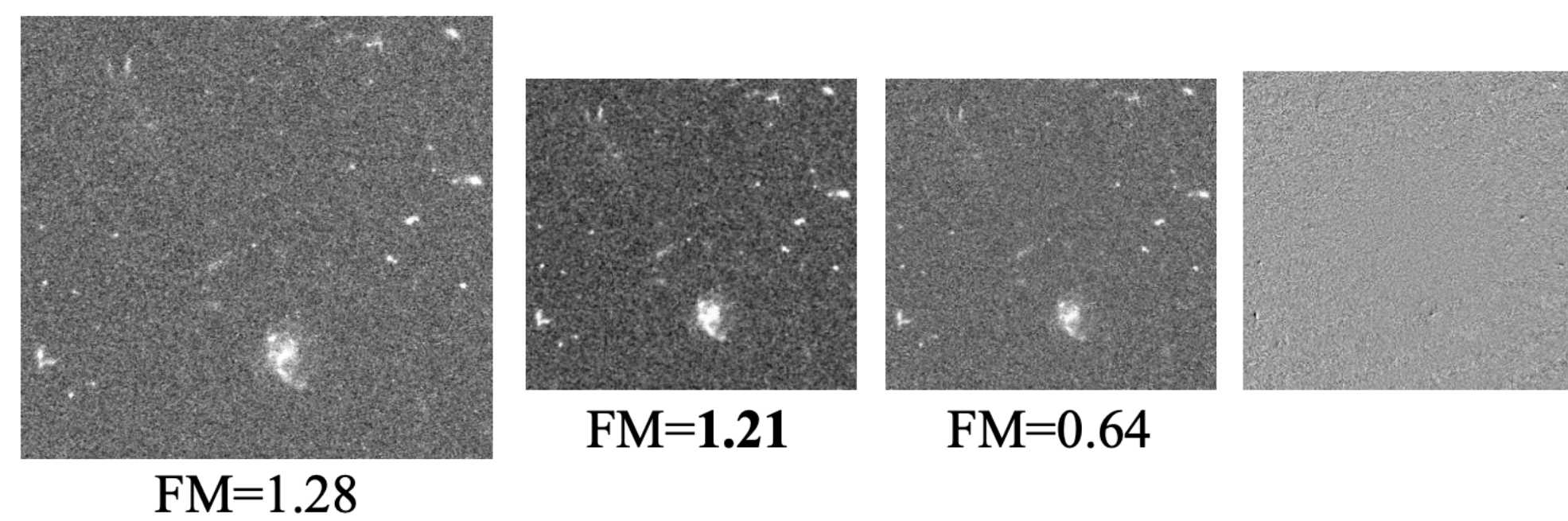
1. Flux Inconsistency
2. Object-Crop Setting
3. Insufficient Data Diversity

(Q&A)

What is flux? In astronomy, flux refers to the total light intensity of celestial bodies (such as stars).

Why should it be consistent? The true telescope should measure the total flux of the same celestial body consistently at different resolutions.

Examples of different downsampling



FM: Flux mean, means the average of the overall flux within this figure.

Interested in our work or joining the team?

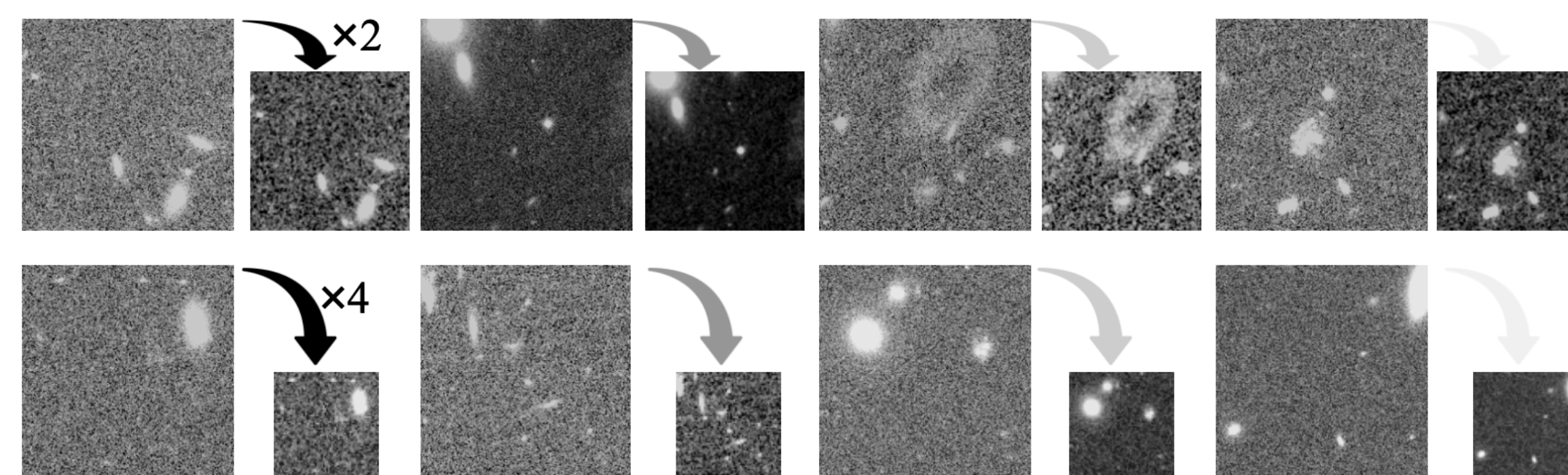
We'd love to chat! Please contact us at: 12guocheng@gmail.com or luyan@pjlabor.org.cn

STAR Overview

A large-scale astronomical SR dataset containing 54,738 flux-consistent star field image pairs covering wide celestial regions.

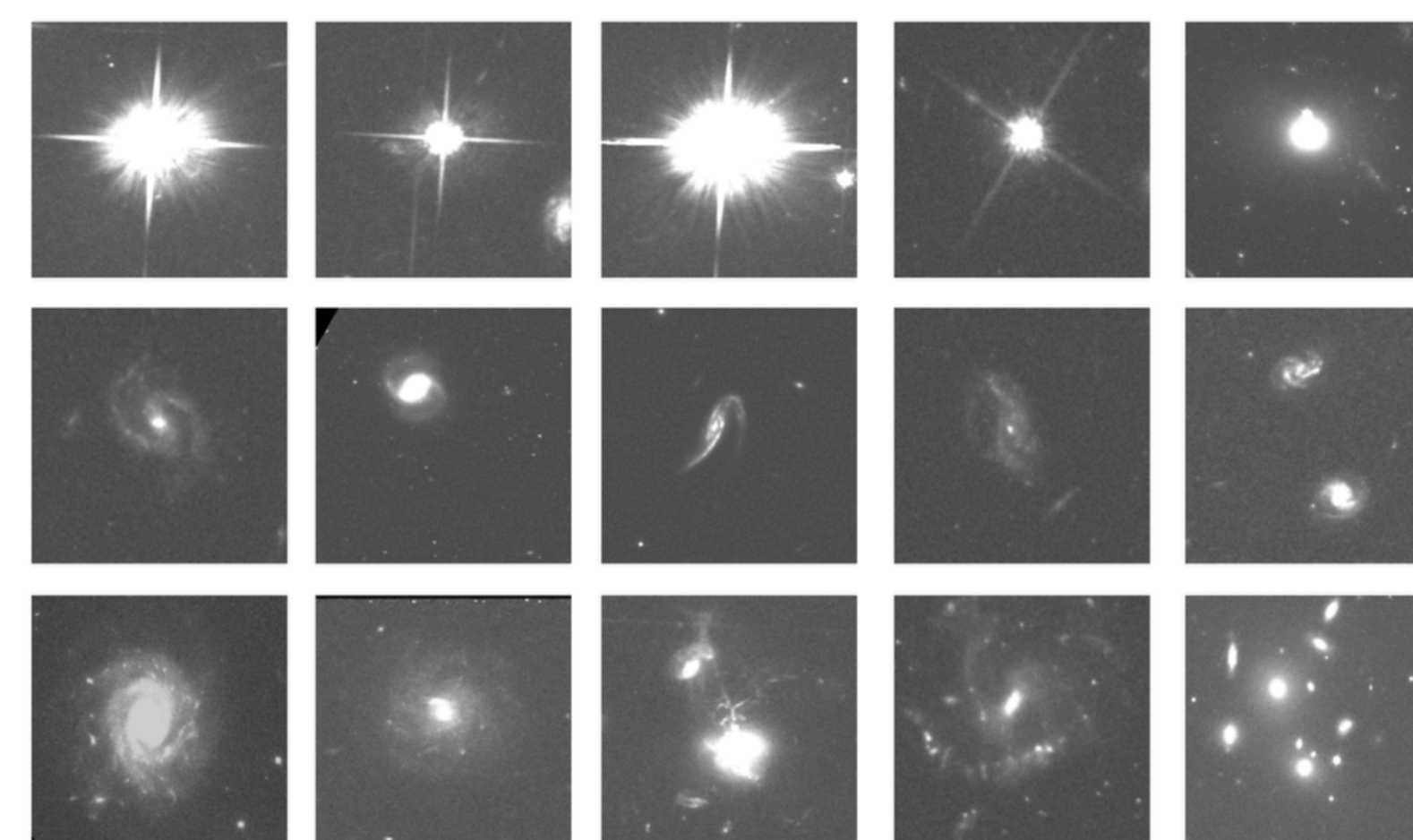
Dataset	Size	Type	Downsampling	Multiple Celestial	Flux Consistency
AstroSR [11]	2000	Galaxy	×	×	×
QQ Shan et al [12]	9383	Galaxy	×	×	×
W Song et al [16]	1597	Solar	✓	×	×
DiffLense [17]	2880	Galaxy	×	×	×
ZJ Luo et al [18]	17000	Galaxy	×	×	×
WJ Li et al [23]	14604	Galaxy	✓	×	×
STAR	54738	Star field	✓	✓	✓

STAR's samples

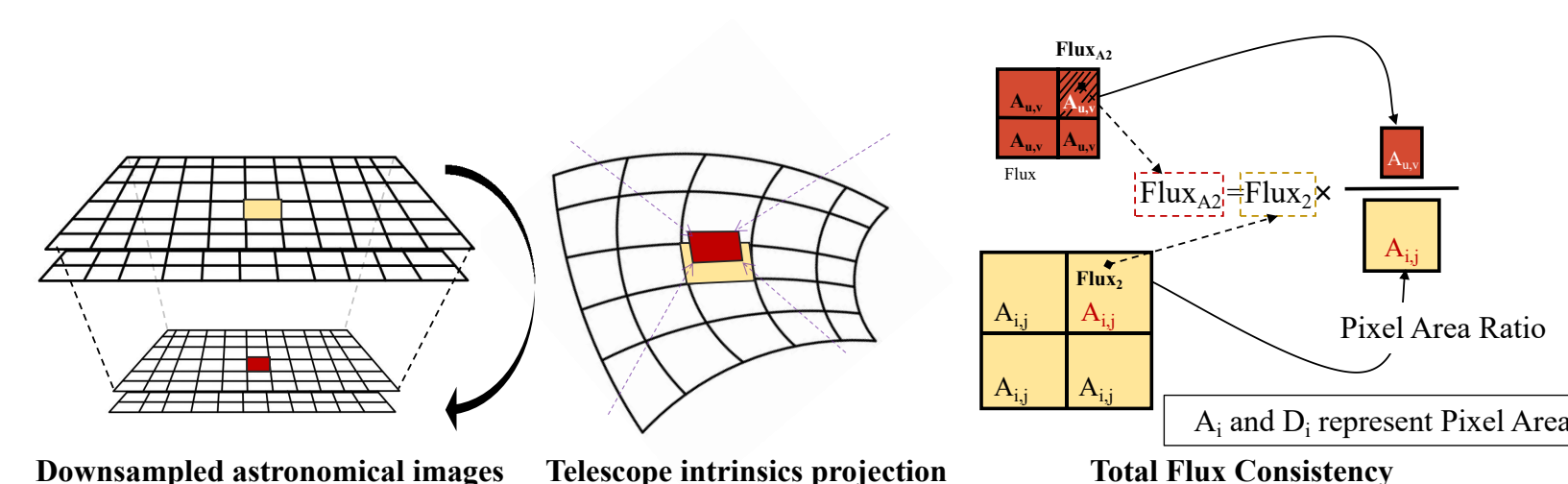


A selection of patches from the STAR dataset, showcasing its diversity.

The examples include **crowded stellar fields**, **regions with interacting galaxies**, and **complex nebulae**, demonstrating the variety of astronomical scenes available.

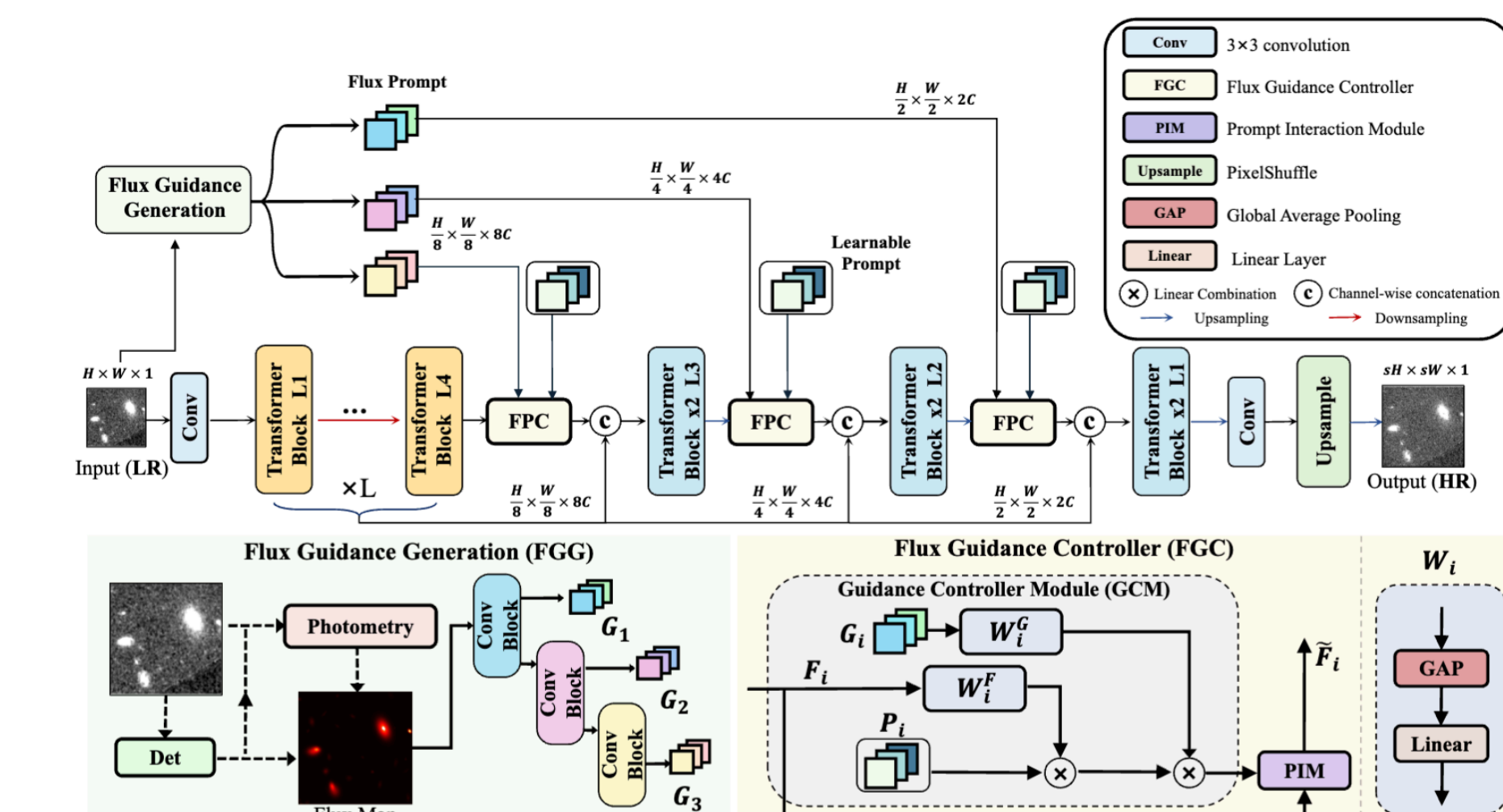


Flux Consistent downsampling process



Methods

We propose a **Flux Consistency Loss (FCL)** for training and a novel **Flux Error (FE)** metric to evaluate the model's physical fidelity.

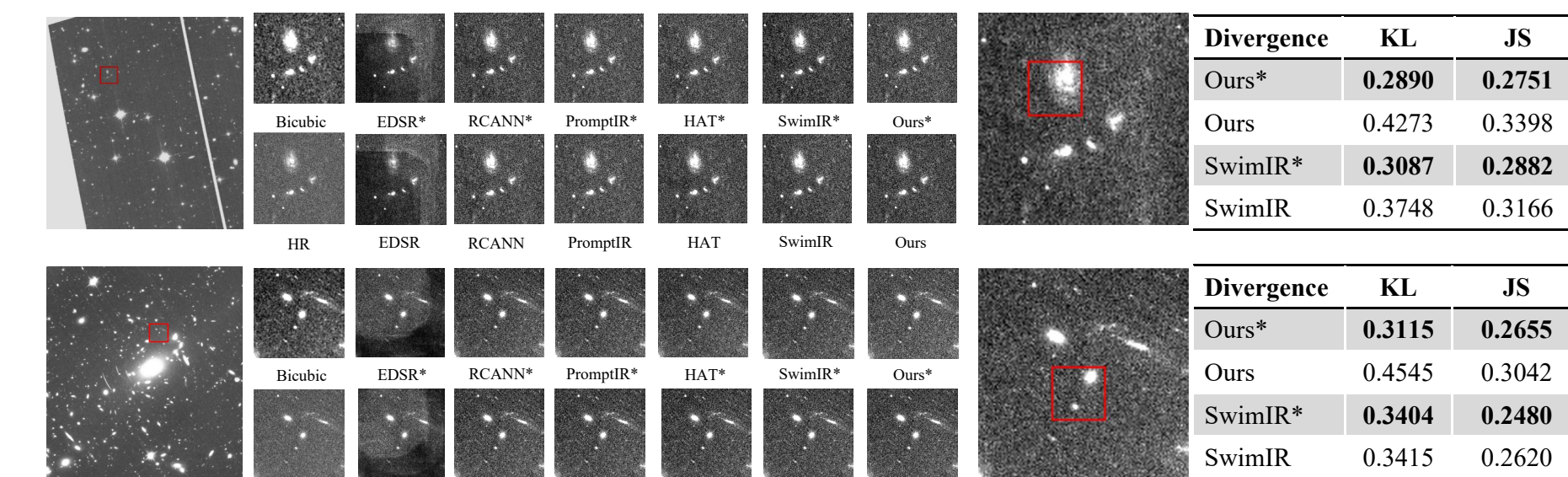


Experiments

Performance comparison of different methods under $\times 2$ and $\times 4$ super-resolution. Evaluation metrics include **PSNR**, **SSIM**, and **Flux Error**.

Scale	Metric	Bicubic	EDSR [8]	RealESRGAN [34]	RCAN [32]	SwinIR [26]	HAT [29]	FISR (ours)
$\times 2$	PSNR $\uparrow$	28.6021	38.8156	36.8363	36.3703	37.2205	37.2501	37.8779
	SSIM $\uparrow$	0.6842	0.8054	0.8225	0.8240	0.8286	0.8295	0.8311
	FE $\downarrow$	4.9418	1.4623	7.3632	0.9237	0.813	0.7636	0.5739
$\times 4$	PSNR $\uparrow$	26.0518	33.8736	34.3725	34.9823	34.6655	34.9142	35.1788
	SSIM $\uparrow$	0.6005	0.7201	0.7223	0.7263	0.7258	0.7276	0.7266
	FE $\downarrow$	7.8733	1.3841	4.0782	1.0550	1.0657	1.0256	1.0125

Result Visualization



Red boxes mark areas for computing KL and JS divergence between predictions and ground truth with (*) are trained using FCL.

Ablation experiment for each module

	FG-Modules	Flux cues	FCL	PSNR↑	SSIM↑	FE↓
1 st				37.7715	0.8288	0.7022
2 nd			✓	37.8570	0.8283	0.6389
3 rd	✓			37.7101	0.8286	0.7572
4 th	✓		✓	37.8681	0.8301	0.7467
5 th	✓	✓		37.8454	0.8302	0.6527
6 th	✓	✓	✓	37.8779	0.8311	0.5739